LABORATORY REPORT

Redcliffe labs

Patient NAME	**: Mr Girish Veerwani**		
DOB/Age/Gender	: 44 Y 10 M 20 D/Male	Report STATUS	: Final Report
Patient ID / UHID	: 13478686/RCL13070353	Barcode NO	: 27470520
Referred BY	: Self	Sample Type	: Whole blood EDTA
Sample Collected	: Aug 12, 2025, 07:07 AM	Report Date	: Aug 12, 2025, 01:01 PM.

NABL M(EL)T LABS NABL-M(EL)T-03061

CM-HC06

Complete Blood Count (CBC)

Test Description	Value(s)	Unit(s)	Reference Range
RBC Parameters			
Hemoglobin *colorimetric*	14.7	g/dL	13.0 - 17.0
RBC Count *Electrical impedance*	5.2	10^6/µl	4.5 - 5.5
PCV *Calculated*	43.9	%	40 - 50
MCV *Calculated*	84.1	fl	83 - 101
MCH *Calculated*	28.2	pg	27 - 32
MCHC *Calculated*	33.6	g/dL	31.5 - 34.5
RDW (CV) * *Calculated*	12	%	11.6 - 14.0
RDW-SD * *Calculated*	37.5	fl	35.1 - 43.9
WBC Parameters			
TLC *Electrical impedance and microscopy*	8.3	10^3/µl	4 - 10
Differential Leucocyte Count			
Neutrophils	63.2	%	40-80
Lymphocytes	28	%	20-40
Monocytes	3.5	%	2-10
Eosinophils	4.5	%	1-6
Basophils	0.8	%	<2
Absolute Leukocyte Counts ***Calculated***			
Neutrophils.	5.25	10^3/µl	2 - 7
Lymphocytes.	2.32	10^3/µl	1 - 3
Monocytes.	0.29	10^3/µl	0.2 - 1.0
Eosinophils.	0.37	10^3/µl	0.02 - 0.5
Basophils.	0.07	10^3/µl	0.02 - 0.5
Platelet Parameters			
Platelet Count *Electrical impedance and microscopy*	286	10^3/µl	150 - 410
Mean Platelet Volume (MPV) * *Calculated*	**8**	fL	9.3 - 12.1
PCT * *Calculated*	0.2	%	0.17 - 0.32

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Test Description	Value(s)	Unit(s)	Reference Range
PDW * *Calculated*	10.5	fL	8.3 - 25.0
P-LCR * *Calculated*	**14.6**	%	18 - 50
P-LCC * *Calculated*	**42**	10^9/L	44 - 140
Mentzer Index * *Calculated*	16.17	%	> 13

Interpretation:
CBC provides information about red cells, white cells and platelets. Results are useful in the diagnosis of anemia, infections, leukemias, clotting disorders and many other medical conditions.

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Sample Collected	: Aug 12, 2025, 07:07 AM	Report Date	: Aug 12, 2025, 01:34 PM.

Test Description	Value(s)	Unit(s)	Reference Range

Erythrocyte Sedimentation Rate (ESR)

ESR - Erythrocyte Sedimentation Rate *MODIFIED WESTERGREN*	6	mm/hr	0 - 10

Interpretation:
ESR is also known as Erythrocyte Sedimentation Rate. An ESR test is used to assess inflammation in the body. Many conditions can cause an abnormal ESR, so an ESR test is typically used with other tests to diagnose and monitor different diseases. An elevated ESR may occur in inflammatory conditions including infection, rheumatoid arthritis ,systemic vasculitis, anemia, multiple myeloma , etc. Low levels are typically seen in congestive heart failure, polycythemia ,sickle cell anemia, hypo fibrinogenemia , etc.

Reference- Dacie and lewis practical hematology

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Test Description	Value(s)	Unit(s)	Reference Range

HbA1C (Glycosylated Haemoglobin)

Glycosylated Hemoglobin (HbA1c) *HPLC*	5.3	%	< 5.7
Estimated Average Glucose	105.41		

Interpretation:
Interpretation For HbA1c% As per American Diabetes Association (ADA)

Reference Group	HbA1c in %
Non diabetic adults >=18 years	<5.7
At risk (Prediabetes)	5.7 - 6.4
Diagnosing Diabetes	>= 6.5
Therapeutic goals for glycemic control	Age > 19 years Goal of therapy: < 7.0 Age < 19 years Goal of therapy: <7.5

Note:
1. Since HbA1c reflects long term fluctuations in the blood glucose concentration, a diabetic patient who is recently under good control may still have a high concentration of HbA1c. Converse is true for a diabetic previously under good control but now poorly controlled.
2. Target goals of < 7.0 % may be beneficial in patients with short duration of diabetes, long life expectancy and no significant cardiovascular disease. In patients with significant complications of diabetes, limited life expectancy or extensive co-morbid conditions, targeting a goal of < 7.0 % may not be appropriate.

Comments :
HbA1c provides an index of average blood glucose levels over the past 8 - 12 weeks and is a much better indicator of long term glycemic control as compared to blood and urinary glucose determinations ADA criteria for correlation between HbA1c & Mean plasma glucose levels.

HbA1c(%)	Mean Plasma Glucose (mg/dL)	HbA1c(%)	Mean Plasma Glucose (mg/dL)
6	126	12	298
8	183	14	355
10	240	16	413

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NABL M(EL)T LABS NABL-M(EL)T-03061

Liver Function Test (LFT)

Test Description	Value(s)	Unit(s)	Reference Range
Bilirubin Total *Photometric*	0.7	mg/dL	0.2 - 1.2
Bilirubin Direct * *Diazo Reaction*	0.2	mg/dL	0.0 - 0.5
Bilirubin Indirect * *Calculation (T Bil - D Bil)*	0.5	mg/dL	0.1 - 1.0
SGOT/AST *IFCC without P5P*	28.9	U/L	5 - 34
SGPT/ALT *IFCC without P5P*	42.5	U/L	0 to 55
SGOT/SGPT Ratio *	0.68	-	-
Alkaline Phosphatase *IFCC*	97	U/L	40 - 150
Total Protein *Biuret*	7.2	g/dL	6.4 - 8.3
Albumin *BCG*	4.8	gm/dL	3.8 - 5.0
Globulin * *Calculation (T.P - Albumin)*	2.4	g/dL	2.3 - 3.5
Albumin :Globulin Ratio * *Calculation (Albumin/Globulin)*	2	-	1.0 - 2.1
Gamma Glutamyl Transferase (GGT) * *Photometric*	26	U/L	12 - 64

Interpretation:
The liver filters blood, metabolizes nutrients, detoxifies harmful substances, and produces blood clotting proteins. Liver cells contain enzymes that facilitate these functions. When cells are damaged, enzymes leak into the blood, detectable through blood tests.

Key enzymes tested:

1. **AST (SGOT):** may indicate tissue injury / damage in muscles or liver.
2. **ALT (SGPT):** Primarily in the liver. Elevated ALT and AST suggest liver damage.
3. **Alkaline Phosphatase & GGT:** Linked to bile production and flow. Elevated levels may indicate bile flow issues related to the liver, gallbladder, or bile ducts.

Blood proteins, **albumin and globulin**, are essential for growth, development, and health.

1. **Low protein:** May indicate bleeding, liver disorders, malnutrition, or agammaglobulinemia.
2. **High protein (Hyperproteinemia):** Often due to dehydration or increased protein production.
3. **Low albumin:** Caused by poor diet, kidney, or liver disease.
4. **High albumin:** Usually due to severe dehydration.

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Kidney Function Test (KFT)

Test Description	Value(s)	Unit(s)	Reference Range
Blood Urea *Urease*	24.3	mg/dL	19.04 - 44.08
Bun * *Urease*	11.36	mg/dL	8.9 - 20.6
Creatinine *Photometric*	**0.7**	mg/dL	0.72 - 1.25
eGFR (CKD-EPI)	115.86	ml/min/1.73 sq m	Normal Or High: >= 90 Mild Or Decrease: 60-89 Mild To Moderate Decrease: 45-59 Mild To Severe Decrease: 30-44 Severe Decrease: 15-29 Kidney Failure: < 15
Bun/Creatinine Ratio * *Calculated*	16.23		12 - 20
Urea / Creatinine Ratio * *Calculated*	34.71		25.68- 42.8
Uric Acid *Uricase*	5.1	mg/dL	3.5 - 7.2
Calcium Serum *Arsenazo III*	8.8	mg/dL	8.4 - 10.2
Phosphorus *Photometric*	3.2	mg/dL	2.3 - 4.7
Sodium *Potentiometric*	145	mmol/L	136 - 145
Potassium *Potentiometric*	**3.2**	mmol/L	3.5 - 5.1
Chloride *Potentiometric*	106	mmol/L	98 - 107

Interpretation:
Kidney function tests is a collective term for a variety of individual tests and proceduresthat can be done toevaluate how well the kidneys are functioning. Many conditions can affect the ability of the kidneys to carryout their vital functions. Somelead to a rapid (acute) decline in kidney functionothers lead to a gradual (chronic) declineinfunction. Both result in a buildup of toxic waste subst done on urine samples, as well as on blood samples. A number of symptoms may indicate a problem with your kidneys. These include : high blood pressure,blood in urine frequent urges to urinate,difficulty beginning urination,painful urination,swelling in the hands and feet due to a buildup of fluids in the body. A single symptom may not mean something serious. However, when occurring simultaneously, these symptoms suggest that your kidneys are not working properly. Kidney function tests can help determine the reason. Electrolytes are present in the human body and the balancing act of the electrolytes in our bodies is essential for normal function of our cells and organs. There has to be a balance.Ionized calcium this test if you have signs of kidney or parathyroid disease. The test may also be done to monitor progress and treatment of these diseases.
"eGFR test is applicable for patients aged 18 years or more."

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Test Description	Value(s)	Unit(s)	Reference Range
Lipid Profile			
Total Cholesterol *Enzymatic - Cholesterol Oxidase*	160	mg/dL	<200
Triglycerides *Colorimetric - Lip/Glycerol Kinase*	**170**	mg/dL	<150
HDL Cholesterol *Accelerator Selective Detergent*	**36.2**	mg/dL	>40
Non HDL Cholesterol * *Calculated*	123.8	mg/dL	<130
LDL Cholesterol * *Calculated*	89.8	mg/dL	<100
V.L.D.L Cholesterol * *Calculated*	**34**	mg/dL	< 30
Chol/HDL Ratio * *Calculated*	4.42	Ratio	3.5 - 5.0
HDL/ LDL Ratio * *Calculated*	**0.4**	Ratio	0.5 - 3.0 Borderline : 3.1 - 6.0 High : > 6.0
LDL/HDL Ratio * *Calculated*	2.48	Ratio	-

Interpretation:
Lipid level assessments must be made following 9 to 12 hours of fasting, otherwise assay results might lead to erroneous interpretation. NCEP recommends of 3 different samples to be drawn at intervals of 1 week for harmonizing biological variables that might be encountered in single assays.

National Lipid Association Recommendations (NLA-2014)	Total Cholesterol (mg/dL)	Triglyceride (mg/dL)	LDL Cholesterol (mg/dL)	Non HDL Cholesterol (mg/dL)
Optimal	<200	<150	<100	<130
Above Optimal			100-129	130 - 159
Borderline High	200-239	150-199	130-159	160 - 189
High	>=240	200-499	160-189	190 - 219
Very High	-	>=500	>=190	>=220

HDL Cholesterol	
Low	High
<40	>=60

Risk Stratification for ASCVD (Atherosclerotic Cardiovascular Disease) by Lipid Association of India.

Risk Category	A. CAD with > 1 feature of high risk group
Extreme risk group	B. CAD with >1 feature of very high risk group of recurrent ACS (within 1 year) despite LDL-C <or = 50 mg/dl or poly vascular disease

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Test Description	Value(s)	Unit(s)	Reference Range

Very High Risk	1.Established ASCVD 2.Diabetes with 2 major risk factors of evidence of end organ damage 3. Familial Homozygous Hypercholesterolemia
High Risk	1. Three major ASCVD risk factors 2. Diabetes with 1 major risk factor or no evidence of end organ damage 3. CHD stage 3B or 4. 4 LDL >190 mg/dl 5. Extreme of a single risk factor 6. Coronary Artery Calcium - CAC > 300 AU 7. Lipoprotein a >/= 50 mg/dl 8. Non stenotic carotid plaque
Moderate Risk	2 major ASCVD risk factors
Low Risk	0-1 major ASCVD risk factors
Major ASCVD (Atherosclerotic cardiovascular disease) Risk Factors	
1. Age >/=45 years in Males & >/= 55 years in Females	3. Current Cigarette smoking or tobacco use
2. Family history of premature ASCVD	4. High blood pressure
5. Low HDL	

Newer treatment goals and statin initiation thresholds based on the risk categories proposed by Lipid Association of India in 2020.

Risk Group	Treatment Goals		Consider Drug Therapy	
	LDL-C (mg/dl)	Non-HDL (mg/dl)	LDL-C (mg/dl)	Non-HDL (mg/dl)
Extreme Risk Group Category A	<50 (Optional goal <OR = 30)	<80 (Optional goal <OR = 60)	>OR = 50	>OR = 80
Extreme Risk Group Category B	>OR = 30	>OR = 60	> 30	> 60
Very High Risk	<50	<80	>OR = 50	>OR = 80
High Risk	<70	<100	>OR = 70	>OR = 100
Moderate Risk	<100	<130	>OR = 100	>OR = 130
Low Risk	<100	<130	>OR = 130*	>OR = 160

*** After an adequate non-pharmacological intervention for at least 3 months.**

References : Management of Dyslipidaemia for the Prevention of Stroke : Clinical practice Recommendations from the Lipid Association of India. Current Vascular Pharmacology,2022,20,134-155.

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Iron Studies

Test Description	Value(s)	Unit(s)	Reference Range
Iron *Ferene*	**58**	µg/dL	65 - 175
TIBC,(Total Iron Binding Capacity) *Calculated*	**244**	µg/dL	250 - 450
UIBC *Ferene*	186	µg/dL	69 - 240
Transferrin Saturation *Method :Derived from IRON and TIBC values*	23.77	%	-

Interpretation:
Increased levels due to iron ingestion or ineffective erythropoiesis.Decreased levels due to infection, inflammation, malignancy, menstruation and Fe deficiency.Needs to be taken into consideration with TIBC. Transferrin Saturation:- Low level Transferrin Saturation can indicate iron deficiency, erythropoiesis, infection, or inflammation. High level Transferrin Saturation can indicate recent ingestion of dietary iron,ineffective erythropoiesis,haemochromatosis or liver disease.High TIBC, UIBC, or transferrin usually indicates iron deficiency, but they are also increased in pregnancy and with the use of oral contraceptives. Low TIBC, UIBC, or transferrin may occur if someone has:Hemochromatosis, Certain types of anemia due to accumulated iron,Malnutrition,kidney disease that causes a loss of protein in urine.

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Test Description	Value(s)	Unit(s)	Reference Range

High Sensitivity C-Reactive Protein (Hs-CRP)

HIGHLY SENSITIVE C-REACTIVE PROTEIN (hs-CRP) *immunoturbidimetric*	**6.6**	mg/L	< 1.00

Interpretation:

Cardio CRP In mg/L	Cardiovascular Risk
<1	Low
1-3	Average
3-10	High
>10	Persistent elevation may represent Non cardiovascular inflammation

Note: To assess vascular risk, it is recommended to test hsCRP levels 2 or more weeks apart and calculate the average

Comments:
High sensitivity C Reactive Protein (hsCRP) significantly improves cardiovascular risk assessment as it is a strongest predictor of future coronary events. It reveals the risk of future Myocardial infarction and Stroke among healthy men and women, independent of traditional risk factors. It identifies patients at risk of first Myocardial infarction even with low to moderate lipid levels. The risk of recurrent cardiovascular events also correlates well with hsCRP levels. It is a powerful independent risk determinant in the prediction of incident Diabetes.

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Sample Collected	: Aug 12, 2025, 07:08 AM	Report Date	: Aug 12, 2025, 02:24 PM.

Test Description	Value(s)	Unit(s)	Reference Range

Vitamin B12 / Cyanocobalamin

Vitamin - B12 *CMIA*	199	pg/mL	187 - 883

Interpretation:
Low Values are a sign of a vitamin B12 deficiency. People with this deficiency are likely to have or develop symptoms.
Causes of vitamin B12 deficiency include:Not enough vitamin B12 in diet (rare except with a strict vegetarian diet), Diseases that cause malabsorption (for example, celiac disease and Crohn's disease), Lack of intrinsic factor, Above normal heat production (for example, with hyperthyroidism), Pregnancy. Increased vitamin B12 levels are uncommon. Usually excess vitamin B12 is removed in the urine. Conditions that can increase B12 levels include: Liver disease (such as cirrhosis or hepatitis), Myeloproliferative disorders (for example, polycythemia vera and chronic myelocytic leukemia).
Vitamin B12: Low Levels can cause malabsorption, Lack of intrinsic factor, Above normal heat production (for example, with hyperthyroidism), Pregnancy.High Level Liver disease, Myeloproliferative disorders (for example, polycythemia vera and chronic myelocytic leukemia).
1. Out of 140 healthy indian population, 91% of Vitamin B 12 concentrations was at lower level: 59.00 pg/ml and upper level: 700.00 pg/ml

"Patients on Biotin supplement may have interference in some immunoassays. Ref: Arch Pathol Lab Med—Vol 141, November 2017. With individuals taking high dose Biotin (more than 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended."

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Test Description	Value(s)	Unit(s)	Reference Range
Vitamin D 25 Hydroxy			
Vitamin D 25 - Hydroxy *CMIA*	**20.5**	ng/mL	Deficiency : <30 ng/mL

Interpretation:
25-Hydroxy vitamin D represents the main body reservoir and transport form. Mild to moderate deficiency is associated with Osteoporosis / Secondary Hyperparathyroidism while severe deficiency causes Rickets in children and Osteomalacia in adults. Prevalence of Vitamin D deficiency is approximately >50% specially in the elderly. This assay is useful for diagnosis of vitamin D deficiency and Hypervitaminosis D. It is also used for differential diagnosis of causes of Rickets & Osteomalacia and for monitoring Vitamin D replacement therapy.

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Test Description	Value(s)	Unit(s)	Reference Range

Thyroid Profile Total

Triiodothyronine (T3) *CMIA*	123.4	ng/dL	35 - 193
Total Thyroxine (T4) *CMIA*	9.8	µg/dL	4.87 - 11.2
Thyroid Stimulating Hormone (Ultrasensitive) *CMIA*	1.3	µIU/mL	0.35 - 4.94

Interpretation:

Pregnancy	Refrence Range TSH
1st Trimester	0 .1 - 2.5
2nd Trimester	0.2 - 3.0
3rd Trimester	0.3 - 3.0

Clinical Use:
1. Diagnose Hypothyroidism & Hyperthyroidism
2. Monitor T4 therapy
3. Measure subnormal TSH levels

Increased TSH: Primary hypothyroidism, Subclinical hypothyroidism, TSH-dependent hyperthyroidism, Thyroid hormone resistance
Decreased TSH: Graves' disease, Autonomous thyroid hormone secretion, TSH deficiency

Thyroid malfunction (hyper or hypo) affects T3 & T4 levels. Pituitary or hypothalamic issues also influence thyroid activity.

1. Primary Hypothyroidism: High TSH levels.
2. Secondary/Tertiary Hypothyroidism: Low TSH levels.
3. Euthyroid Sick Syndrome: Abnormal thyroid test results due to non-thyroidal illnesses (NTI).

TBG levels are stable in healthy individuals but may be altered by pregnancy, estrogens, androgens, steroids, or glucocorticoids, causing inaccurate T3 & T4 readings.

TSH	T4	T3	Interpretation
High	Normal	Normal	Mild (subclinical) hypothyroidism
High	Low	Low Or Nomral	Hypothyroidism
Low	Normal	Normal	Mild (subclinical) hyperthyroidism
Low	High Or Normal	High Or Normal	Hyperthyroidism
Low	Low Or Normal	Low Or Normal	Nonthyroidal illness; pituitary (secondary) hypothyroidism
Normal	High	High	Thyroid hormone resistance syndrome (a mutation in the thyroid hormone receptor decreases thyroid hormone function)

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ilac-MRA MC-5280

Test Description	Value(s)	Unit(s)	Reference Range

Magnesium

Magnesium,Serum *Colorimetric*	1.01	mmol/L	0.66 - 1.07

Interpretation:

Age	Ref Range in mmol/L
Newborn, 2 to 4 days	0.62 - 0.91
5 months to 6 years	0.70 - 0.95
6 to 12 years	0.70 - 0.86
12 to 20 years	0.70 - 0.91
Adult	0.66 - 1.07
60 - 90 years:	0.66 - 0.99
> 90 years:	0.70 - 0.95

Moderate or severe magnesium deficiency is usually due to losses of magnesium from gastrointestinal tract or kidneys as in vomiting and diarrhoea in former and alcohol, diabetes mellitus (osmotic diuresis), loop diuretics(furosemide) and aminoglycoside antibiotics in latter. Symptomatic hypermagnesemia is almost always caused by excessive intake with concomitant renal failure, thereby decreasing the ability of the kidneys to excrete excess magnesium.

Lactate Dehydrogenase (LDH), Serum

LDH:Lactate Dehydrogenase *IFCC*	213	U/L	up to 250

Interpretation:
1-Marked elevations in Lactate Dehydrogenase (LDH) activity can be observed in megaloblastic anemia, untreated pernicious anaemia, Hodgkin's disease, abdominal and lung cancers, severe shock, and hypoxia.
2-Moderate to slight increases in LDH levels are seen in myocardial infarction (MI), pulmonary infarction, pulmonary embolism, leukemia, hemolytic anemia, infectious mononucleosis, progressive muscular dystrophy (especially in the early and middle stages of the disease), liver disease, and renal disease.
3-In liver disease, elevations of LHD are not as great as the increases in aspartate amino transferase (AST) and alanine aminotransferase (ALT).
4-Increased levels of the enzyme are found in about one third of patients with renal disease, especially those with tubular necrosis or pyelonephritis. However, these elevations do not correlate well with proteinuria or other parameters of renal disease On occasion a raised LDH level may be the only evidence to suggest the presence of a hidden pulmonary embolus.

Caution:
1-Red blood cells contain much more lactate dehydrogenase (LDH) than serum. A hemolyzed specimen is not acceptable. LDH activity is one of the most sensitive indicators of in vitro hemolysis. Causes can include transportation via pneumatic tube, vigorous mixing, or traumatic venipuncture.

*** End Of Report ***

Pending Report To Follow - Urine R/M

(*) Parameter(s) are outside the scope of tests recognized under the NABL M(EL)T Scheme.

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1	2	3	4	5	6	7	8	9	10

Not Likely — Very Likely

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